Data generation or arrival rates always come in bursts. Sometimes, data rates may be only a few bits per second (bps) and abruptly go as high as a few megabits per second (Mbps). In telephony, transport of voice signal is sent over a pre-established link between the caller and the recipient. The link established between the two remains dedicated for the entire period of conversation. No other potential user can use the link then.

Any pre-established link between the caller and the recipient can carry data of assigned/designed bit rate, which is fixed. Let us say, a link is of 1 kilobits per seconds (kbps). If such pre-established and dedicated link of fixed data rate is made to carry bursty data, two major engineering problems come up. First, when data arrival or generation is only a few bps, link utilisation is poor.

In the above example, if data arrival rate is one bps, utilisation (defined as data arrival rate/data service or link rate) will be 1bps/1kbps

This is extremely poor. From an engineering point of view, the solution is not at all cost-effective.

Second, when data arrival rate in burst is very high, say, 1Mbps, the link of 1kbps will not be able to carry data. Data will be lost in this case. In real life, it is like bursty traffic on road. When traffic is low, road utility is poor. When traffic is high, a traffic jam occurs.

In telephonic or voice communication, communication from the caller to the recipient, and vice-versa, is almost symmetric—50 per cent both ways, or 40 per cent on one end and 60 per cent on the other, and so on.

Unlike voice, data communication is mostly asymmetric. Flow of data in one direction is often more than that in other direction. It is like when you click on a link to download a file. If the dedicated fixed-rate telephonic link is used to carry data, due to asymmetric nature of data, the link utility in one direction will be poor. Whereas, in the other direction it may be overloaded, causing loss of data. Data transport is, thus, unsuitable over a dedicated fixed-rate telephonic link.

In voice/telephonic communication, the caller and the recipient are intelligent entities, aka, human beings.

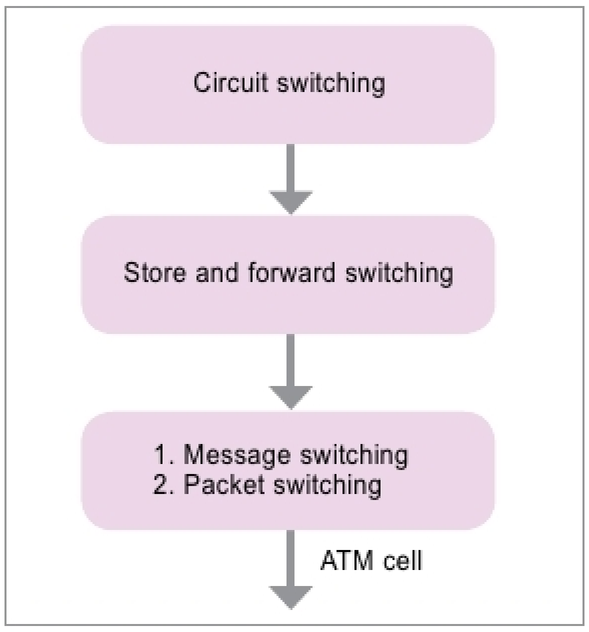

Fig. 1: Virtual circuit services and datagram services

Whereas, in data communication, communicating entities are machines, computers or other electronics devices. Due to human intelligence and perception, if there is any error or confusion in voice communication, the same is corrected by the communicating agencies. For example, when the caller says, "It is too big," but it reaches the recipient as, "It is to big," human perception and intelligence allow the recipient to correct it.

Correction of error in data communication is not possible, as end devices are machines. Thus, data transport is delicate. If a data in binary form, say, 10000000, from a source reaches with a single error at the left most bit to a receiver, it will be received as 00000000. This issue is addressed using due data correction strategy in networks, which is absent in telephonic networks.

No voice communication is perfect without being instantaneous. Telephonic communication is no exception. For example, a voice message saying, "I will go home tomorrow" having a three-day delay or latency is pointless.

That is not the case with data. It may not be a big deal if your bank statement reaches your e-mail address after three days of requisition. Transport of data with zero delay is a quality and not a necessity. Whereas, voice communication with zero delay is nature-specific and characteristically a necessity.

Data is insensitive to time, which means data transport may tolerate a